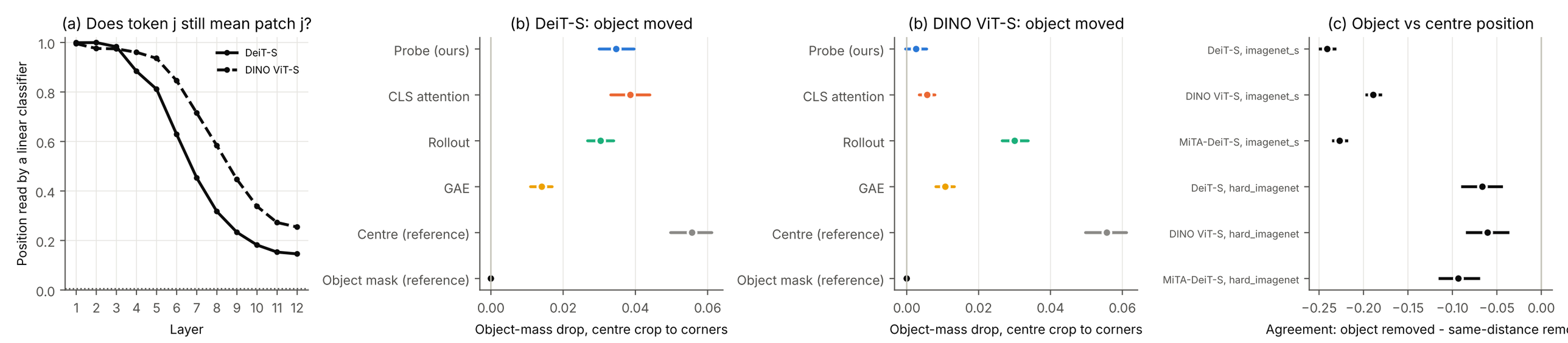
(a) Does token j still mean patch j?
Position read by a linear classifier
1.0
0.8
0.6
0.4
0.2
0.0
1 2 3 4 5 6 7 8 9 10 11 12
Layer
DeiT-S
DINO ViT-S
(b) DeiT-S: object moved
Probe (ours)
CLS attention
Rollout
GAE
Centre (reference)
Object mask (reference)
0.00 0.02 0.04 0.06
Object-mass drop, centre crop to corners
(b) DINO ViT-S: object moved
Probe (ours)
CLS attention
Rollout
GAE
Centre (reference)
Object mask (reference)
0.00 0.02 0.04 0.06
Object-mass drop, centre crop to corners
(c) Object vs centre position
DeiT-S, imagenet_s
DINO ViT-S, imagenet_s
MiTA-DeiT-S, imagenet_s
DeiT-S, hard_imagenet
DINO ViT-S, hard_imagenet
MiTA-DeiT-S, hard_imagenet
−0.25 −0.20 −0.15 −0.10 −0.05 0.00
Agreement: object removed - same-distance